Review of the deep-dive notes against primary evidence

The supplied notes add useful derivations to Jiaxuan's blog synthesis. We preserve both documents and maintain this review separately. This pass checks the architecture and hyperparameter claims most relevant to the current questions; the remaining numerical claims from later reports have not all been independently audited.

Checked claims and qualifications

Claim in the notes	Primary evidence	How to use it
A single width/depth progression is required	AI2 Table 1 uses 12/16/16/16 layers in the ladder and 32/40 in targets, with changes to head dimensions and MLP ratio	A uniform family is a useful design choice. Successful published ladders can depart from exact geometric similarity; test those departures empirically
µP transfers LR across width	Tensor Programs V §7.4 transfers from width 256 to 4,096 at 32 blocks	Distinguish base LR from matrix-specific actual LR. Specify initialization and readout conventions too
µP does not handle batch or horizon	Tensor Programs V §6.1 investigates transfer beyond width with caveats	Width theory does not supply universal batch/horizon optima; empirical transfer beyond width is not ruled out
Depth transfer fails for Transformers	Tensor Programs VI studies a particular class of infinite-depth parameterizations and distinguishes one-layer from multi-layer residual blocks	Do not interpret this as a universal impossibility theorem. Audit residual scaling and retune at changed depths
Batch-size scaling has a consensus exponent 0.4–0.57	Step Law v3 uses 0.571; Power Lines v2 obtains 0.383 with tuned decay and other exponents with fixed decay	These are conditional fits. Record batch units, data, optimizer, and decay-tuning policy
Optimal AdamW τ stays constant as D changes	Power Lines v2 instead finds $\tau_* \approx 1.084(D/N)^{-0.527}$	Constant τ supports scaling decay with batch at fixed N, D; it is not the cross-TPP optimum
Repetition decay is approximately 15.4, rather than 4	Muennighoff et al., Appendix A, Eq. 14 reports $R_D^* = 15.387756$ and $R_N^* = 5.309743$	The notes' correction is supported for that fit. Four epochs is not the fitted decay constant
Every candidate-comparison ladder requires exhaustive per-point tuning	AI2 transfers a prescribed recipe; µTransfer deliberately avoids large-model sweeps	Tune enough for the claim being made. A fixed-recipe comparison and a fully tuned frontier are different estimands

Sources: AI2 task ladder, v2, §2.1/Table 1; Tensor Programs V, v2; Tensor Programs VI; Step Law v3; Power Lines v2, §§2–3/Table 5; data-constrained scaling, v5, Appendix A.

Two details that matter when using the derivations

The Skaling outer exponent preserves the **algebraic form** of the compute-optimal allocation for positive k under $C = 6ND$. Refitting the coupled model can change A, B, α, β, and therefore change the numerical recommendation. "Same allocation formula" does not mean the two separately fitted models must recommend the same size.

The floor E in a fitted loss surface is an estimated parameter. Calling it the exact entropy of the underlying text distribution is an idealization. Uncertainty in E can strongly affect equivalent-compute ratios near the floor; propagate it into the decision rather than interpreting a fitted value as known physical truth.

Evidence standard for the next deep dives

For each practical rule, record the exact model configurations, what was held fixed, data/token horizon, optimizer and parameterization, measured result, figure/table locator, and transfer limitation. A table of chosen hyperparameters establishes a recipe used in practice; it does not establish optimality or a causal scaling law without the corresponding sweep.

The architecture evidence guide applies this standard to the AI2 ladder, μP, fixed-depth optimizer comparisons, batch size, and weight decay.